

Development of a Dynamic Ride-Sharing and Carpooling Platform

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Abstract: The project titled Development of a Dynamic Ride-Sharing and Carpooling Platform focuses on creating a web-based transportation system called Smart Ride Sharing. The platform allows drivers and passengers traveling in the same direction to share rides conveniently. Drivers can publish their trip details such as source, destination, travel date, and available seats, while passengers can search for suitable rides and book seats according to their travel requirements. The main objective of the Smart Ride Sharing website is to make travel more efficient, economical, and accessible. The Smart Ride Sharing platform includes several core modules such as user registration and authentication, ride posting and booking, and route-based ride matching. The system also implements dynamic fare calculation based on the distance traveled by each passenger. To enhance usability, the platform provides booking confirmations, real-time notifications, and a secure online payment system that ensures smooth and reliable transactions between drivers and passengers. Additionally, the platform contains an administrative dashboard to monitor ride activities, manage users, and track payments within the system. A review and rating feature is also included to maintain transparency and trust between drivers and passengers. By promoting shared transportation through the Smart Ride Sharing website, the system aims to reduce traffic congestion, minimize travel costs, and contribute to a more sustainable and eco-friendly transportation environment.

Index terms: Smart Ride Sharing, Carpooling, Ride Booking System, Dynamic Fare Calculation, Route-Based Matching, Spring Boot, Secure Payment Gateway, Real-Time Notifications, User Authentication, Sustainable Transportation.

I. INTRODUCTION

Rapid urbanization and the increasing number of private vehicles have resulted in traffic congestion, higher transportation costs, environmental pollution, and inefficient utilization of available vehicle capacity. Ride-sharing and carpooling have emerged as effective solutions to these challenges by enabling multiple travelers with similar routes to share a single vehicle. Such systems not only reduce travel expenses and fuel consumption but also contribute to sustainable transportation by minimizing traffic density and carbon emissions. With the widespread adoption of web and mobile technologies, intelligent ride-sharing platforms have gained significant attention as a convenient and efficient mode of transportation management.

This paper presents the Development of a Dynamic Ride-Sharing and Carpooling Platform, implemented as a full-stack web application named Smart Ride Sharing. The proposed system enables drivers to publish ride information and allows passengers to search, book, and manage rides through a secure online environment. The platform incorporates features such as user authentication, ride

management, dynamic fare calculation, payment processing, notifications, reviews, and administrative monitoring. The system was developed using React.js, Java Spring Boot, MySQL, REST APIs, JPA Hibernate ORM, and JWT Authentication to ensure scalability, security, and efficient system performance. The proposed solution aims to enhance transportation accessibility, optimize resource utilization, and provide a reliable platform for shared mobility services.

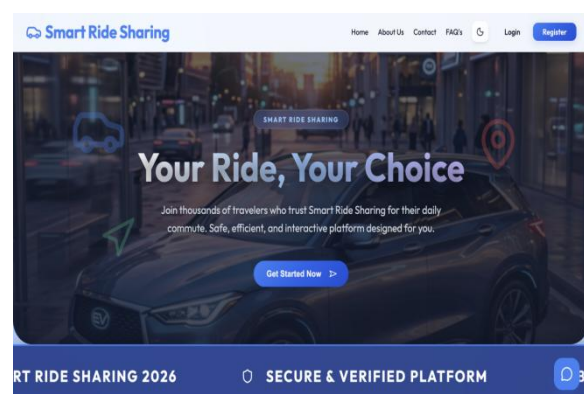


Figure 1: Home Page

II. LITERATURE REVIEW

The concept of ride-sharing and carpooling has gained significant attention in recent years as an effective solution for reducing traffic congestion, transportation costs, and environmental pollution. Several researchers have proposed technology-based transportation systems that connect drivers and passengers through web and mobile applications. These systems primarily focus on improving travel convenience, optimizing vehicle utilization, and encouraging shared mobility. The integration of modern web technologies, cloud services, and digital payment solutions has further enhanced the efficiency and accessibility of ride-sharing platforms.

Previous studies have highlighted the importance of secure user authentication, route matching, real-time communication, and online payment processing in transportation management systems. Many existing ride-sharing applications provide functionalities such as ride posting, ride searching, booking management, fare estimation, and user feedback mechanisms. Researchers have also emphasized the role of GPS-based navigation, notification services, and review systems in improving user trust and overall platform reliability. These features contribute to better coordination between drivers and passengers while ensuring a seamless travel experience.

Several commercial ride-sharing platforms have successfully demonstrated the benefits of shared transportation; however, certain limitations still exist. Some systems face challenges related to user verification, ride coordination, booking transparency, payment security, and efficient administrative monitoring. In addition, many existing solutions offer limited customization and may not adequately address the requirements of specific user groups or regional transportation needs. These challenges indicate the need for more secure, scalable, and user-centric ride-sharing platforms.

To address these limitations, the proposed Smart Ride Sharing System integrates secure JWT-based authentication, ride management, booking services, dynamic fare calculation, payment processing, notification management, review and rating features, and an administrative dashboard within a unified web-based platform. The system is designed using modern full-stack technologies to provide improved security, maintainability, scalability, and user convenience. By combining multiple transportation management functionalities into a single application, the proposed solution aims to enhance the overall efficiency and effectiveness of ride-sharing services.

III. SYSTEM ARCHITECTURE

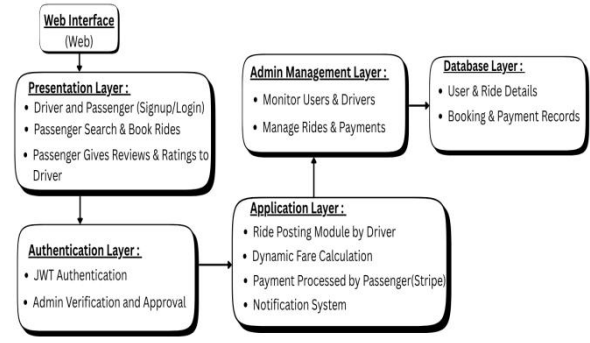


Figure 2: Layered Architecture of the Smart Ride Sharing System.

The Smart Ride Sharing System was developed using a layered architecture to ensure efficient communication between users, application services, administration, and data storage components. As illustrated in **Figure 2**, the system consists of five layers: Presentation Layer, Authentication Layer, Application Layer, Admin Management Layer, and Database Layer. This design improves system organization, security, scalability, and maintainability by assigning specific responsibilities to each layer.

1. Presentation Layer

The Presentation Layer provides the interface through which drivers and passengers interact with the platform. Users can register, log in, search rides, book rides, and submit reviews using responsive web pages developed with React.js, HTML, CSS, and JavaScript. This layer focuses on delivering a smooth and user-friendly experience.

2. Authentication Layer

The Authentication Layer manages user verification and access control within the system. JWT-based authentication is used to validate user identities and maintain secure login sessions. It ensures that only authorized users can access protected features and perform permitted actions.

3. Application Layer

The Application Layer handles the main functionalities of the platform, including ride posting, ride booking, fare calculation, payment processing, and notification management. It processes user requests, applies business rules, and coordinates communication between the frontend and database components. This layer serves as the core operational unit of the system.

4. Admin Management Layer

The Admin Management Layer allows administrators to oversee platform activities through a centralized dashboard. It provides features for monitoring users, rides, bookings, payments, and system performance. This layer helps maintain operational control, transparency, and secure management of the application.

5. Database Layer

The Database Layer stores and manages all application data, including user information, ride details, bookings, payments, and reviews. MySQL and JPA Hibernate ORM were used to support efficient data storage and retrieval. This layer ensures data integrity, reliability, and secure handling of information across the platform.

IV. METHODOLOGY

The development of the Smart Ride Sharing System followed a structured full-stack web application approach to ensure efficient implementation, security, and scalability. The methodology involved designing the user interface, developing backend services, managing database operations, implementing secure authentication mechanisms, and integrating online payment functionality. Modern technologies such as React.js, Java Spring Boot, MySQL, JWT Authentication, and Stripe Payment Gateway were utilized to develop a reliable and user-friendly ride-sharing platform.

1. Frontend Development

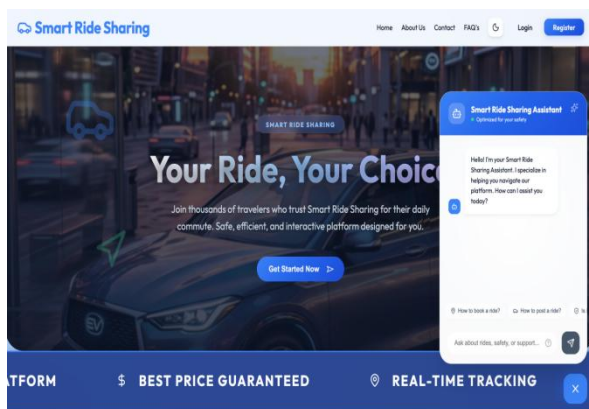


Figure 3: Chatbot and Home Page

As shown in Figure 3. The frontend of the Smart Ride Sharing System was developed using React.js, HTML, CSS, and JavaScript to create responsive and interactive web pages. Various interfaces such as the Home Page, Login Page, Registration Page, Ride

Search Page, Booking Page, Dashboard, FAQ's Page, Contact Page, and Chatbot were implemented to provide smooth user interaction. React.js components were used to improve code reusability and maintainability, while responsive design techniques ensured compatibility across desktop, tablet, and mobile devices.

2. Backend Development

The backend was implemented using Java Spring Boot to manage application logic and server-side processing. Multiple modules were developed to handle user registration, ride posting, ride booking, fare calculation, notifications, review management, and administrative operations. REST APIs were created to enable communication between frontend components and backend services. The backend architecture followed the MVC design pattern, which improved code organization and simplified maintenance.

3. Database Implementation

MySQL was used as the database management system for storing and managing application data. Several tables were designed to maintain user information, ride details, vehicle records, bookings, payments, notifications, reviews, and support queries. JPA Hibernate ORM was implemented to establish object-relational mapping between Java classes and database tables, reducing manual query handling and improving data management efficiency. Proper relationships and constraints were defined to ensure data consistency and reliability.

4. Authentication and Security

JWT (JSON Web Token) authentication was integrated to provide secure user login and session management. During authentication, user credentials were verified, and secure access tokens were generated for authorized users. Token expiration functionality was implemented to automatically terminate inactive sessions and redirect users to the login page when required. Role-based access control was also applied to manage permissions for drivers, passengers, and administrators, ensuring secure access to system resources.

5. Payment Integration

As shown in Figure 4. Online payment functionality was implemented using the Stripe Payment Gateway to support secure transaction processing within the platform. Passengers can complete ride payments through a secure card-based payment system after booking confirmation. The payment module records

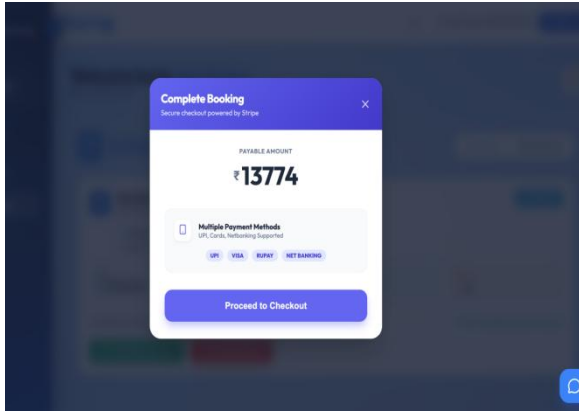


Figure 4: Stripe Payment Gateway

transaction details, maintains payment history, and updates booking status accordingly. This integration improved transaction security, payment transparency, and overall user convenience within the Smart Ride Sharing System.

V. IMPLEMENTATION AND RESULTS

The implementation phase involved integrating all major modules of the Smart Ride Sharing System into a unified web application. Technologies such as React.js, Java Spring Boot, MySQL, JPA Hibernate ORM, JWT Authentication, and Stripe Payment Gateway were used to develop a secure and efficient platform. Various functionalities including user management, ride booking, payment processing, notifications, reviews, and administrative monitoring were implemented and tested successfully. The completed system provided a seamless experience for drivers, passengers, and administrators while ensuring reliable performance and secure data management.

1. User Registration and Login

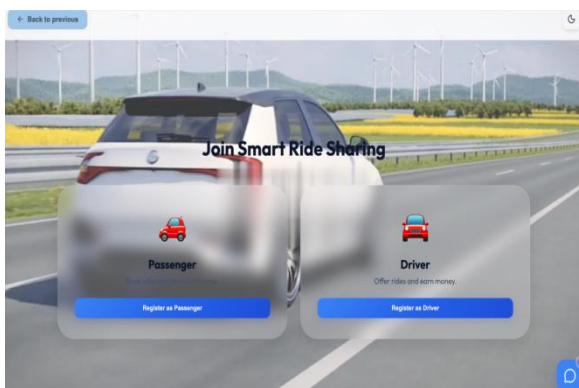


Figure 5: Registration Page

As shown in Figure 5 and 6. The User Registration and Login module was implemented to provide

secure access to the Smart Ride Sharing System. Separate registration options were provided for drivers and passengers, where users were required to submit personal details, address information, educational information, and identity verification documents such as Aadhaar Card or PAN Card. Login functionality was secured using JWT authentication, allowing authorized users to access the platform based on their assigned roles. Successful implementation of this module ensured secure account management and protected access to application resources.

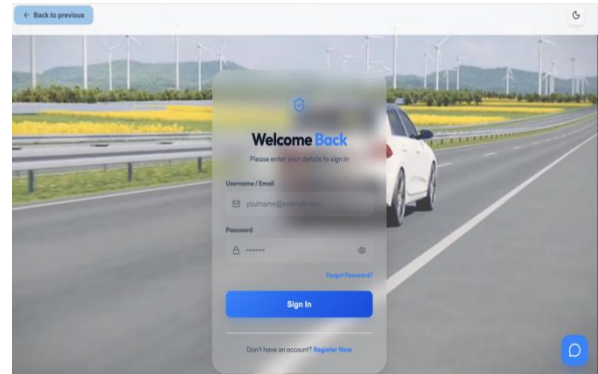


Figure 6: Login Page

2. Ride Posting and Booking

The Ride Posting and Booking module enabled drivers to create and manage rides by entering travel details such as source location, destination, travel date, available seats, vehicle information, and fare details. Passengers could search available rides, apply route-based filters, select preferred rides, and reserve seats through the booking system. Booking records were maintained within the database and updated dynamically according to ride availability. This module successfully streamlined ride-sharing operations and improved coordination between drivers and passengers as shown in Figure 7.

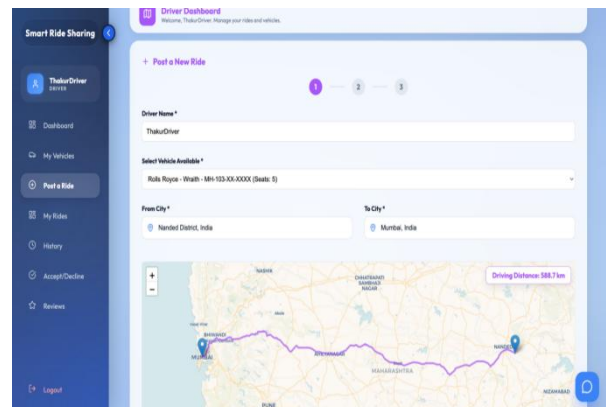


Figure 7: Driver Ride Posting Page

3. Fare Calculation

The Fare Calculation module was developed to determine ride costs automatically based on travel distance and ride-related parameters. The system calculated fares dynamically whenever a ride was created or booked, reducing manual calculations and improving pricing accuracy. Fare information was displayed to passengers before booking confirmation to ensure transparency. This functionality simplified payment processing and enhanced user trust by providing clear and consistent fare estimation throughout the booking process.

4. Notification System, Review and Rating System

The Notification System was implemented to keep users informed about important ride-related activities. Notifications were generated for booking confirmations, ride updates, payment status changes, ride reminders, and cancellation events. Along with notifications, a Review and Rating System was developed to allow passengers and drivers to share feedback after ride completion. Users could submit ratings and comments based on their travel experience, helping improve service quality, accountability, and trust within the platform. These features enhanced communication and user engagement throughout the ride-sharing process.

5. Admin Dashboard, Driver and Passenger Dashboards

Dedicated dashboards were developed for administrators, drivers, and passengers to provide role-specific functionalities. The Admin Dashboard enabled centralized monitoring of users, rides, bookings, payments, reviews, and overall system activities. The Driver Dashboard allowed drivers to manage ride postings, monitor bookings, update ride status, and view passenger information. The Passenger Dashboard provided access to ride searches, booking history, notifications, payment details, and submitted reviews. The successful implementation of these dashboards improved usability, operational management, and overall efficiency of the Smart Ride Sharing System.

VI. ADVANTAGES OF PROPOSED SYSTEM

1. Reduces Traffic Congestion

The Smart Ride Sharing System encourages multiple passengers traveling in the same direction to share a single vehicle. This reduces the number of vehicles on the road, helping to decrease traffic congestion in

urban areas and improve overall transportation efficiency.

2. Minimizes Transportation Costs

By allowing passengers to share travel expenses, the system helps reduce individual transportation costs. Users can access affordable travel options while drivers can partially recover fuel and travel expenses through ride-sharing arrangements.

3. Enhances Security and Reliability

JWT-based authentication, role-based access control, and secure payment processing help protect user information and prevent unauthorized access. These security features create a trustworthy platform for drivers and passengers to interact safely.

4. Improves User Convenience

The platform provides ride posting, ride searching, online booking, payment integration, notifications, reviews, and dashboard management within a single application. These features simplify the complete ride-sharing process and improve the overall user experience.

5. Supports Environmental Sustainability

By promoting shared transportation, the system helps reduce fuel consumption and unnecessary vehicle usage. Fewer vehicles on the road contribute to lower carbon emissions and support environmentally friendly transportation practices.

VII. FUTURE SCOPE

The Smart Ride Sharing System can be further enhanced by incorporating advanced technologies to improve efficiency, user convenience, and transportation management. One potential improvement is the implementation of AI-based ride matching, which can automatically connect drivers and passengers with similar routes, schedules, and travel preferences. This would optimize vehicle utilization, reduce waiting time, and improve the overall ride-sharing experience.

The system can also be extended with GPS-based real-time tracking to provide live vehicle location updates and accurate route monitoring. Such functionality would improve passenger safety, ride transparency, and travel coordination. Additionally, integrating machine learning algorithms can enable personalized ride recommendations, demand prediction, and intelligent route optimization based on user behavior and historical travel patterns.

Another important enhancement is the development of a dedicated mobile application for Android and iOS platforms. A mobile-based solution would provide greater accessibility, real-time notifications, location-based services, and improved user engagement. These future advancements would make the Smart Ride Sharing System more intelligent, scalable, and suitable for modern smart transportation environments.

CONCLUSION

The Smart Ride Sharing System was successfully developed as a secure and efficient web-based platform that connects drivers and passengers for convenient ride-sharing services. The application integrates essential functionalities such as user registration, authentication, ride posting, ride booking, dynamic fare calculation, payment processing, notifications, reviews, and administrative monitoring within a single system. Technologies including React.js, Java Spring Boot, MySQL, JWT Authentication, and JPA Hibernate ORM were utilized to build a scalable, reliable, and user-friendly solution. By encouraging shared transportation, the platform helps reduce traffic congestion, optimize vehicle utilization, lower travel costs, and improve overall commuting convenience. The successful implementation of these features demonstrates the effectiveness of modern web technologies in addressing real-world transportation challenges.

The project also provided valuable practical experience in full-stack web application development, database management, API integration, authentication handling, testing, and debugging. Throughout the development process, software engineering principles and industry-standard practices were applied to ensure quality, security, and maintainability. The experience enhanced technical knowledge, problem-solving abilities, and understanding of real-world project development workflows. Furthermore, the system offers opportunities for future enhancements such as AI-based ride matching, GPS-enabled tracking, machine learning recommendations, and mobile application support. Overall, the project achieved its intended objectives and serves as a strong foundation for developing advanced smart transportation solutions in the future.

REFERENCES

- [1] D. Zhang et al., "Carpooling service for large-scale taxicab networks," *ACM Trans. Sensor Netw.*, vol. 12, no. 3, p. 18, 2016.
- [2] S. Ma, Y. Zheng, and O. Wolfson, "T-share: A large-scale dynamic taxi ridesharing service," in *Proc. IEEE 29th Int. Conf. Data Eng.*, Apr. 2013, pp. 410–421.
- [3] uberPOOL Together. Accessed: Oct. 11, 2018. [Online]. Available: <https://www.uber.com/en-PK/ride/uberpool/>
- [4] J. Li and V. H. Allan, "Balancing taxi distribution in a city-scale dynamic ridesharing service: A hybrid solution based on demand learning," in *Proc. IEEE Int. Smart Cities Conf. (ISC)*, Sep. 2020, pp. 1–8.
- [5] N. Agatz, A. L. Erera, M. W. P. Savelsbergh, and X. Wang, "Dynamic ride-sharing: A simulation study in metro Atlanta," *Proc. Social Behav. Sci.*, vol. 17, pp. 532–550, Jan. 2011.
- [6] Z. Dong and M. Leng, "Managing on-demand ridesharing operations: Optimal pricing decisions for a ridesharing platform," *Int. J. Prod. Econ.*, vol. 232, Feb. 2021, Art. no. 107958.
- [7] S. Y. Amirkiaee and N. Evangelopoulos, "Why do people rideshare? An experimental study," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 55, pp. 9–24, May 2018.
- [8] M. Zhu, X.-Y. Liu, and X. Wang, "An online ride-sharing path-planning strategy for public vehicle systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 2, pp. 616–627, Feb. 2019.
- [9] M. Zhu, X.-Y. Liu, F. Tang, M. Qiu, R. Shen, W. Shu, and M.-Y. Wu, "Public vehicles for future urban transportation," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 12, pp. 3344–3353, Dec. 2016.
- [10] C. Certoma, F. Corsini, and F. Rizzi, "Crowdsourcing urban sustainability. Data, people and technologies in participatory governance," *Futures*, vol. 74, pp. 93–106, Nov. 2015.
- [11] NYC Taxi and Limousine Commission. Taxi of Tomorrow Survey Results. Accessed: Oct. 13, 2018. [Online]. Available: http://www.nyc.gov/html/tlc/downloads/pdf/tot_survey_results_02_10_11.pdf
- [12] Car-Pooling Declines as Driving Becomes Cheaper. Accessed: Nov. 21, 2017. [Online]. Available: <http://www.nytimes.com/interactive/2011/01/29/us/20110129-nat-CARPOOL.html>
- [13] M. Furuhashi, M. Dessouky, F. Ordóñez, M.-E. Brunet, X. Wang, and S. Koenig, "Ridesharing: The state-of-the-art and future directions," *Transp. Res. B, Methodol.*, vol. 57, pp. 28–46, Nov. 2013.
- [14] M. M. Solomon, "Algorithms for the vehicle routing and scheduling problems with time window constraints," *Oper. Res.*, vol. 35, no. 2, pp. 254–265, 1987.
- [15] B. Cao, L. Alarabi, M. F. Mokbel, A. Basalamah, S. Engineering, and S. Arabia, "SHAREK: A scalable dynamic ride sharing system," in *Proc. 16th IEEE Int. Conf. Mobile Data Manage.*, Pittsburgh, PA, USA, Jun. 2015, pp. 234–240.
- [16] Y.-L. Chen, K.-F. Ssu, and Y.-J. Chang, "Real-time transfers for improving efficiency of ridesharing services in the environment with connected and self-driving vehicles," in *Proc. Int. Comput. Symp. (ICS)*, Dec. 2020, pp. 165–170.

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